Independent macOS Security Research Initiative

Executive Summary

This proposal seeks funding to advance AI-assisted macOS security research through the development and evaluation of an open-source platform known as MacOS Audit Agent. The project combines artificial intelligence with human-led security research to improve detection, auditing, and defensive capabilities on modern Apple Silicon systems.

Researcher Background

I am an independent security researcher with more than six years of experience in cybersecurity, security research, vulnerability analysis, and defensive security technologies. My work focuses on practical security tooling, operating system internals, AI-assisted research workflows, and defensive security automation.

Problem Statement

While AI-assisted development has rapidly improved productivity, macOS security remains comparatively underserved. Many AI systems appear optimized for Windows and Linux workflows, often lacking deep familiarity with Apple-specific security models, Apple Silicon architecture, launchd persistence, Endpoint Security APIs, system telemetry, and other macOS-specific defensive concepts.

The goal of this project is to close that gap through structured experimentation, dataset collection, continuous evaluation, and development of practical defensive tools.

Why Apple Silicon Research Matters

Apple Silicon has rapidly become a major computing platform used by consumers, developers, educators, researchers, and enterprises. As adoption grows, the need for effective defensive tooling also increases. Research into AI-assisted security analysis on ARM64 and Apple Silicon platforms can help improve defensive outcomes while identifying areas where current AI systems require additional training, validation, or workflow improvements.

Project Description

MacOS Audit Agent is an open-source security platform that functions as both a local auditing framework and a host-based intrusion detection system. The project focuses on identifying suspicious activity, persistence mechanisms, privilege escalation opportunities, misconfigurations, and emerging threats while generating actionable remediation guidance.

Research Objectives

• Advance AI-assisted security analysis for macOS.

• Improve defensive tooling for Apple Silicon systems.

• Evaluate GPT-5.5 and Codex-assisted security workflows.

• Identify AI strengths, limitations, and failure modes.

• Produce openly documented defensive methodologies.

Methodology

Data will be collected daily through testing, system snapshots, telemetry review, development logs, and outcome tracking. Each AI-generated recommendation will be evaluated and documented. Successful and unsuccessful approaches will be recorded to produce a measurable understanding of AI-assisted security research effectiveness.

Monthly Milestones

Month 1: Environment setup, hardware acquisition, baseline measurements.

Month 2: Detection engine improvements and telemetry collection.

Month 3: AI-assisted workflow evaluation and dataset generation.

Month 4: Advanced detection capabilities and validation testing.

Month 5: Documentation, community feedback, and refinement.

Month 6: Final reporting, publication, and deliverables.

Measurable Success Criteria

• Daily dataset collection completed.

• Improvement in detection and auditing coverage.

• Documented AI workflow successes and failures.

• Public or private research report delivered.

• Open-source improvements committed to the project.

Deliverables

Comprehensive research notes, datasets, testing results, AI workflow analysis, platform improvements, technical documentation, and final recommendations.

Availability of Results

Results can be shared privately if required. If public release is permitted, datasets, findings, methodologies, and lessons learned will be documented for the benefit of the broader security community.

Proposed Budget (6-Month Research Period)

Researcher Compensation: $31,200

Apple Silicon Development Hardware: $3,500

Test Devices and Equipment: $2,000

OpenAI/API Usage: $3,000

Conferences and Professional Development: $2,000

Miscellaneous Research Expenses: $2,300

Total Request: $44,000